

Functions of two variables—Differentiability

Here is some technical stuff that you don't need to know for Math 133 but you will find important if you go on to Math 235.

Some stuff from one-variable calculus

First, let's review some one-variable stuff you should already know.

A function can be continuous but not differentiable. For example, the function $y = |x|$ is continuous but is not differentiable at $x = 0$.

A function can be differentiable but have a discontinuous derivative. Consider the function defined by $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. We can calculate $f'(0)$ by going back to the difference-quotient definition of the derivative:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} h \sin(1/h) = 0. \quad (1)$$

For $x \neq 0$ we have $f'(x) = 2x \sin(1/x) - \cos(1/x)$. Now $\lim_{x \rightarrow 0} f'(x)$ doesn't exist (because $\lim_{x \rightarrow 0} \cos(1/x)$ doesn't exist), so we don't have $\lim_{x \rightarrow 0} f'(x) = f'(0)$, so f' exists for all x but is not continuous at $x = 0$.

Functions of two variables—partial derivatives

Now let's look at functions of two variables.

The definition of partial derivative is examinable for Math 133:

Definition: If $z = f(x, y)$ then

$$\frac{\partial z}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \quad (2)$$

if the limit exists (and similarly for $\partial z / \partial y$).

The usual way to calculate $z_x(a, b)$ (here, z_x abbreviates $\partial z / \partial x$) is to use the standard rules of one-variable calculus (sum rule, product rule, etc.) to calculate $\partial z / \partial x$ and then evaluate the result at (a, b) . However, this doesn't always work, and there are times when you have to return to the definition. For example, suppose $z = x^{1/3}y^{1/3}$ and we want $z_x(0, 0)$. We have $\partial z / \partial x = (1/3)x^{-2/3}y^{1/3}$, and we cannot evaluate this at $(0, 0)$ since it gives the indeterminate form $0/0$. Using (2) instead we get

$$\frac{\partial z}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0. \quad (3)$$

Differentiability

Although the notes define partial derivatives (and directional derivatives) no definition is given for the simple term, *differentiable*. Here's a definition (not examinable for Math 133, but important for Math 235).

Definition: The function $z = f(x, y)$ is differentiable means that $\partial z/\partial x$ and $\partial z/\partial y$ exist and

$$\lim_{(h,k) \rightarrow (0,0)} \frac{f(x+h, y+k) - f(x, y) - (\partial z/\partial x)h - (\partial z/\partial y)k}{\sqrt{h^2 + k^2}} = 0 \quad (4)$$

To get some idea of where this formula comes from, consider that the difference-quotient definition in one variable, $f'(x) = \lim_{h \rightarrow 0} ((f(x+h) - f(x))/h)$, can be rewritten as

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x) - hf'(x)}{h} = 0. \quad (5)$$

Formula (4) is what happens when you try to write (5) for two variables.

Note that the function $f(x, y) = x^{1/3}y^{1/3}$ is *not* differentiable at $(0, 0)$. As we have seen, the partial derivatives exist, and are zero, at $(0, 0)$, so, using (4), it comes down to a question of whether

$$\lim_{(h,k) \rightarrow (0,0)} \frac{h^{1/3}k^{1/3}}{\sqrt{h^2 + k^2}} \quad (6)$$

is zero. But if you take $k = h$, the function in (6) becomes $(1/\sqrt{2})h^{-1/3}$, which certainly does not approach zero as $h \rightarrow 0$.

There is a theorem that says that if the partial derivatives are continuous then the function is differentiable. Note that in the $z = x^{1/3}y^{1/3}$ example, the partial derivatives are not continuous at $(0, 0)$. For example, $z_x(a, a) = (a^{-1/3})/3$ does not approach $z_x(0, 0) = 0$ as $a \rightarrow 0$.

It is possible for a function to be differentiable even if the partial derivatives are not continuous. An example can be based on the one-variable example, $f(x) = x^2 \sin(1/x)$, given earlier. Another example is given by

$$f(x, y) = x^4/(x^2 + y^2) \text{ for } (x, y) \neq (0, 0), f(0, 0) = 0. \quad (7)$$

You can check (by using the definition of partial derivative) that $z_x(0, 0) = z_y(0, 0) = 0$, and that the limit in (4) comes down to $\lim_{(h,k) \rightarrow (0,0)} (h^4/(h^2 + k^2)^{3/2})$, and that this limit is zero because $|h^4/(h^2 + k^2)^{3/2}| \leq |h^4/h^3| = |h|$, so the function is differentiable. But for $(x, y) \neq (0, 0)$ you get $\partial z/\partial x = (2x^4 + 4x^2y^2)/(x^2 + y^2)^2$ which does not approach $z_x(0, 0) = 0$ as $(x, y) \rightarrow (0, 0)$ (see, for example, what happens when $x = y$).

Directional Derivatives

This is how directional derivatives are defined in the notes (and this is examinable for Math 133):

Definition. If $z = f(x, y)$ and $\mathbf{v} = (a, b)$ is a unit vector then the directional derivative of z in the direction of $\mathbf{v}$ is the dot product, $(\partial z/\partial x, \partial z/\partial y) \cdot (a, b)$.

A definition given in lecture is the following:

Definition. If $z = f(x, y)$ and $\mathbf{v} = (a, b)$ is a unit vector then the directional derivative of z in the direction of $\mathbf{v}$ is the value at $t = 0$ of $\frac{d}{dt}f(x + at, y + bt)$.

This is a logical definition, as it measures how fast f changes as you move away from (x, y) in the direction (a, b) . You can get from one definition to the other by the chain rule, but the chain rule is only valid if f is differentiable. If f is not differentiable then the two definitions can give different answers, and in such cases the second definition is the one to use. Here is an example. Let

$$z = f(x, y) = x^3/(x^2 + y^2) \text{ for } (x, y) \neq (0, 0), f(0, 0) = 0. \quad (8)$$

Then to calculate $z_x(0, 0)$ we set $y = 0$, giving $f(x, 0) = x$, differentiate, and evaluate, to get $z_x(0, 0) = 1$. To calculate $z_y(0, 0)$ we set $x = 0$, giving $f(0, y) = 0$, differentiate, and evaluate, to get $z_y(0, 0) = 0$. So we have

$$\left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right) \cdot (a, b) = a \quad (9)$$

On the other hand, we have $f(at, bt) = \frac{a^3}{a^2 + b^2}t = a^3t$, so the directional derivative at $(0, 0)$ in the direction (a, b) is

$$\frac{d}{dt}a^3t = a^3, \quad (10)$$

which is the correct value.